

EV-BMS-100 Battery Management System

Design Document • Rev. C • Ayr Powertrain Systems • Doc. No. DD-BMS-100-C

1. Overview

The EV-BMS-100 is a 96S lithium-ion battery management system for light commercial electric vehicle packs. It provides cell-level voltage and temperature monitoring, passive balancing, contactor control, and a CAN 2.0B interface to the vehicle ECU. The BMS MCU implements over-voltage, under-voltage, over-current, and over-temperature protection independently of the vehicle ECU.

2. System Architecture

Figure 1 — BMS System Block Diagram

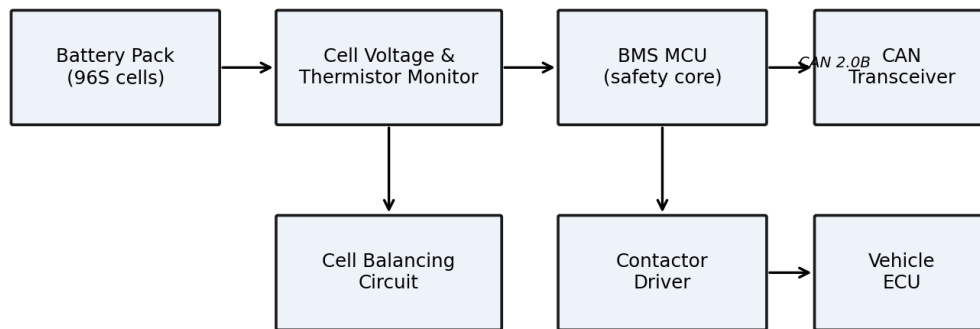


Figure 1 — BMS system block diagram showing the monitoring, balancing, and communication paths between the battery pack and the vehicle ECU.

The Cell Voltage & Thermistor Monitor samples all 96 cell taps and 12 thermistor channels at 10 Hz and reports results to the BMS MCU over an isolated SPI link. The BMS MCU runs the safety state machine and issues balancing commands to the Cell Balancing Circuit and open/close commands to the Contactor Driver.

3. Electrical Specifications

Parameter	Value	Notes
Nominal pack voltage	355.2 V	96S1P, 3.7 V/cell nominal
Max charge current	120 A	continuous, 150 A for 30 s
Max discharge current	320 A	continuous, 450 A for 10 s
Cell balancing current	150 mA	passive, per cell
Overvoltage threshold	4.25 V/cell	latching fault
Undervoltage threshold	2.80 V/cell	latching fault
Isolation voltage	≥ 500 V DC	pack-to-chassis
Quiescent current (sleep)	< 2 mA	measured at 12 V aux supply
Communication protocol	CAN 2.0B, 500 kbps	J1939-style PGNs
Operating temperature range	-30 °C to 65 °C	ambient, enclosure-mounted

4. Mechanical Specifications

Parameter	Value	Notes
Enclosure dimensions	280 x 210 x 65 mm	L x W x H
Weight	2.4 kg	excluding harness
Ingress protection	IP67	connectors mated
Mounting	4x M6 threaded inserts	150 x 100 mm pattern

Figure 2 — BMS Enclosure (rendering)

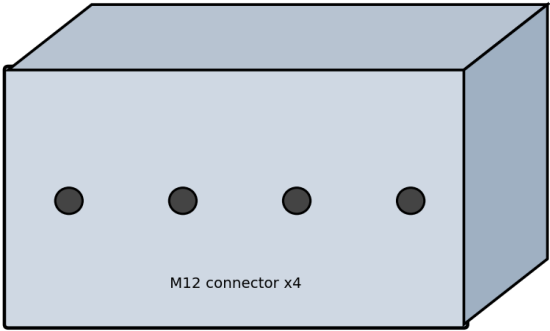


Figure 2 — BMS enclosure rendering showing the four M12 connector ports (pack sense, power, CAN, and auxiliary 12 V).

5. Known Limitations

The current hardware revision (Rev. C) does not support active balancing; balancing energy is dissipated as heat across bleed resistors. A future Rev. D is planned to add active balancing via a flyback topology, which is expected to improve pack-level capacity utilization by 2-3%.